

1. Consider the region R determined by the curves $y = \cosh(x)$, $x = 0$, $y = 0$, and $x = 1$.
a) Sketch a picture of the region R .

- b) Set up, but do not evaluate, an integral which represents the area of the region R .
- c) Suppose that R is revolved around the x -axis to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.
- d) Suppose that R is revolved around the line $x = 2$ to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.

2. Consider the region R determined by the curves $y = \tanh(x)$, $y = \tanh(1)$, $y = \tanh(0.25)$, & $x = 0$.

a) Sketch a picture of the region R .

b) Set up, but do not evaluate, an integral(s) which represents the area of the region R .

c) Suppose that R is revolved around the line $y = 1$ to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.

d) Suppose that R is revolved around the y -axis to create a solid. Setup, but do not evaluate, an integral which represents the volume of the solid.

3. Consider the curve $y = 2 - \cosh(x)$ on $[-1, 1]$.

a) Setup, but do not evaluate, an integral which represents the length of the curve.

b) Suppose the curve is rotated around the x -axis. Setup, but do not evaluate, an integral which represents the surface area of the resulting surface.

4. Consider the curve $y = 2 \sinh(x) - 1$ on $[0, \ln(2)]$.

a) Setup, but do not evaluate, an integral which represents the length of the curve.

b) Suppose the curve is rotated around the x -axis. Why are we not able to use the surface area formula here to help find the surface area for the resulting surface?

c) Suppose that we now consider the curve $y = 2\sinh(x) - 1$ on the interval $\left[\frac{1}{2}, 1\right]$. If this curve is rotated about the x -axis, then setup, but do not evaluate, an integral which represents the surface area of the resulting surface.

d) Suppose that the same curve from part (c) is now revolved about the y -axis. Setup, but do not evaluate, an integral which represents the surface area of the resulting surface.